

Phase Space Debug Report: m036

Automated Pipeline

March 8, 2026

1 Gluing Equations (SnapPy)

Manifold: m036, $n = 4$ tetrahedra, $r = 1$ cusp. Columns ordered as $Z_1, Z'_1, Z''_1, Z_2, Z'_2, Z''_2, Z_3, Z'_3, Z''_3, Z_4, Z'_4, Z''_4$.

Edge equations (C matrix, shape 4×12):

$$C = \begin{bmatrix} 2 & 0 & 1 & 1 & 2 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Note: $n - r = 3$ edge equations are independent; the fourth is their sum (all four are shown as SnapPy returns them).

Cusp equations (row order: μ_0, λ_0):

$$C_{\text{cusp}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 2 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2 Geometric Basis

The pipeline finds $n - r = 3$ independent internal edges. These are classified into **2 hard** edges (carrying fugacity variables η_1, η_2) and **1 easy** edge (whose charge is summed freely, with no fugacity variable).

Easy edge criterion: a basis edge whose 3n-vector has exactly one non-zero entry per tetrahedron, lying in the Z' slot. Such edges contribute only $I_{\Delta}(0, e)$ factors and can be freely summed.

- **Easy Edge Q_3 :** Vector = $[0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0]$
- **Hard Edge Q_1 :** Vector = $[2, 0, 1, 1, 2, 0, 1, 1, 0, 1, 1, 0]$
- **Hard Edge Q_2 :** Vector = $[0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1]$

Since there are 2 hard edges, the refined index carries **two fugacity variables** η_1, η_2 .

3 Neumann-Zagier Matrix

Row structure of $g_{NZ} \in \text{Sp}(8, \mathbb{Q})$ ($2n \times 2n$ with $n = 4$, columns in block order $Z_1, Z_2, Z_3, Z_4, Z_1'', Z_2'', Z_3'', Z_4''$, rows 0 : μ_0 rows 1, 2 : Q_1, Q_2 (hard) row 3 : Q_3 (easy) row 4 : $\frac{\lambda_0}{2}$ rows 5, 6, 7 : $\Gamma_1, \Gamma_2, \Gamma_3$)

g_{NZ} Matrix:

$$g_{NZ} = \begin{bmatrix} 0 & -2 & 0 & 0 & 1 & -2 & -1 & 0 \\ 2 & -1 & 0 & 0 & 1 & -2 & -1 & -1 \\ 0 & 1 & 0 & -1 & 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 & -1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -2 & -2 & 1 & 1 & -3 & -1 & 0 \\ 0 & -2 & 0 & 2 & 0 & -2 & -1 & 0 \end{bmatrix}$$

Affine Shift ν :

$$\nu_X = \begin{bmatrix} 2 \\ 2 \\ -1 \\ -1 \end{bmatrix}, \quad \nu_P = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The rows of ν_X correspond to position rows (rows 0–3: μ_0, Q_1, Q_2, Q_3) and ν_P to momentum rows (rows 4–7: $\lambda_0/2, \Gamma_1, \Gamma_2, \Gamma_3$).

4 Inversion and Local Charges

Inverse Matrix g_{NZ}^{-1} (all-integer, verified $g_{NZ}g_{NZ}^{-1} = I$):

$$g_{NZ}^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 & -1 & -1 & 0 & 1 \\ -1 & 0 & -3 & -2 & 2 & 2 & -1 & -1 \\ 0 & 0 & -1 & -1 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 2 & 0 & -1 \\ 1 & 0 & 2 & 2 & -2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -1 & -2 & 0 & 0 & -1 & 1 \end{bmatrix}$$

The master charge vector is

$$\kappa = [m, 0, 0, 0; e, e_{h_1}, e_{h_2}, e_{\text{easy}}]^T,$$

where m is the cusp meridian charge, the three zeros are internal position charges (always fixed to zero in the summation), e is the cusp longitude/2 charge, $e_{h_1}, e_{h_2} \in \frac{1}{2}\mathbb{Z}$ are the hard-edge momentum charges (carrying fugacities $\eta_1^{2e_{h_1}}, \eta_2^{2e_{h_2}}$), and $e_{\text{easy}} \in \frac{1}{2}\mathbb{Z}$ is the easy-edge momentum charge (summed freely with no fugacity).

The local tetrahedron charges $\Delta = g_{NZ}^{-1} \kappa$:

$$\begin{aligned} \Delta_1 : \quad \tilde{m}_1 &= -e - e_{h_1} + e_{\text{easy}}, & \tilde{e}_1 &= m + 2e_{h_1} - e_{\text{easy}} \\ \Delta_2 : \quad \tilde{m}_2 &= -m + 2e + 2e_{h_1} - e_{h_2} - e_{\text{easy}}, & \tilde{e}_2 &= m - 2e - e_{h_1} + e_{h_2} \\ \Delta_3 : \quad \tilde{m}_3 &= e + e_{h_1} - e_{h_2}, & \tilde{e}_3 &= e_{\text{easy}} \\ \Delta_4 : \quad \tilde{m}_4 &= e_{h_1}, & \tilde{e}_4 &= -e_{h_2} + e_{\text{easy}} \end{aligned}$$

The topological framing phase power $\phi = \vec{m}_{\text{full}} \cdot \nu_P - \vec{e}_{\text{full}} \cdot \nu_X$, where $\vec{m}_{\text{full}} = (m, 0, 0, 0)$ and $\vec{e}_{\text{full}} = (e, e_{h_1}, e_{h_2}, e_{\text{easy}})$. Substituting $\nu_P = (1, 0, 0, 0)$ and $\nu_X = (2, 2, -1, -1)$:

$$\phi = m \cdot 1 - (e \cdot 2 + e_{h_1} \cdot 2 + e_{h_2} \cdot (-1) + e_{\text{easy}} \cdot (-1)) = m - 2e - 2e_{h_1} + e_{h_2} + e_{\text{easy}}.$$

5 Final Summation Formula

The refined index is:

$$I_{m036}(m, e) = \sum_{\substack{e_{h_1}, e_{h_2}, e_{\text{easy}} \\ \in \frac{1}{2}\mathbb{Z}}} \eta_1^{2e_{h_1}} \eta_2^{2e_{h_2}} \cdot (-q^{1/2})^\phi \cdot \prod_{i=1}^4 I_\Delta(\tilde{m}_i, \tilde{e}_i),$$

where $\phi = m - 2e - 2e_{h_1} + e_{h_2} + e_{\text{easy}}$ and the $\tilde{m}_i, \tilde{e}_i$ are given above.

Evaluated output at $(m, e) = (0, 0)$ (truncated at q^4):

$$I = 1$$

$$\begin{aligned} &+ q(-\eta_2^{-1} - 3 - \eta_1\eta_2) \\ &+ q^2(\eta_1^{-1}\eta_2^{-2} + \eta_1^{-1}\eta_2^{-1} - \eta_2^{-1} - 1 + \eta_2 - \eta_1\eta_2 + \eta_1\eta_2^2) \\ &+ q^3(\eta_1^{-1}\eta_2^{-3} + 3\eta_1^{-1}\eta_2^{-2} + \eta_2^{-2} + 3\eta_2^{-1} + 10 + \eta_1 + 3\eta_1\eta_2 + 3\eta_1\eta_2^2 + \eta_1^2\eta_2^2 + \eta_1^2\eta_2^3) \\ &+ q^4(\eta_1^{-1}\eta_2^{-3} + \eta_1^{-1}\eta_2^{-2} - 4\eta_1^{-1}\eta_2^{-1} + \eta_1^{-1} + \eta_2^{-2} + 10\eta_2^{-1} + 14 - 4\eta_2 + 3\eta_1 + 10\eta_1\eta_2 + \eta_1\eta_2^2 + \eta_1^2\eta_2^2 + \eta_1^2\eta_2^3) \end{aligned}$$

Specialisation to 3D index (setting $\eta_1 = \eta_2 = 1$, projecting over all fugacity sectors):

$$I_{3D}(q) = 1 - 5q + q^2 + 27q^3 + 34q^4 + \dots$$

(This matches the direct 3D-index computation via `compute_index_3d_python`.)

Raw key convention (as stored in `RefinedIndexResult`): a monomial $q^k \eta_1^a \eta_2^b$ is stored under key $(2k, 2a, 2b)$, i.e. all exponents doubled to allow half-integer q -powers.

Integrality / half-integrality of summation variables. The charges $e_{h_1}, e_{h_2}, e_{\text{easy}}$ are summed over $\frac{1}{2}\mathbb{Z}$. For a given (m, e) , only those values that make every local charge $\tilde{m}_i, \tilde{e}_i$ simultaneously a half-integer (relative to the SnaPy orientation) contribute. The pipeline enforces this by requiring

$$g_{NZ}^{-1} \kappa \in \left(\frac{1}{2}\mathbb{Z}\right)^{2n}.$$